🛒 Zepto Inventory Data Analysis using SQL

📌 Project Overview

This project analyzes a **Zepto product inventory dataset** using **MySQL**. The objective is to perform data cleaning, exploratory data analysis (EDA), and business-oriented SQL analysis to extract meaningful insights from the dataset.

The project demonstrates the use of SQL concepts such as filtering, aggregation, grouping, sorting, window functions, and conditional statements.

📂 Dataset

The dataset contains product-level information with the following columns:

Column	Description
id	Unique product identifier
Category	Product category
name	Product name
mrp	Maximum Retail Price
discountPercent	Discount offered (%)
discountedSellingPrice	Selling price after discount
availableQuantity	Available inventory
weightInGms	Product weight (grams)
outOfStock	Stock availability
quantity	Product quantity/pack size

🛠️ SQL Concepts Used

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- DISTINCT

- Aggregate Functions (COUNT, SUM, AVG, MAX)
- CASE Statement
- Window Functions (`SUM() OVER()`)
- ALTER TABLE
- LIMIT

📊 Analysis Performed

1. Data Exploration

- Displayed complete dataset
- Counted total number of records
- Added Auto Increment Primary Key (`id`)
- Displayed sample records
- Checked for NULL values

2. Inventory Analysis

- Products in stock vs out of stock
- Product names having multiple SKUs
- Categories containing more than 20 products

3. Pricing Analysis

- Top 20 products with the highest discount percentage
- High-priced products currently out of stock
- Average MRP by category
- Most expensive product in each category
- Price per gram calculation

4. Revenue Analysis

- Estimated inventory value by category using:

`discountedSellingPrice × availableQuantity`

5. Product Classification

- Classified products into:
- Low Weight (<1000 g)
- Medium Weight (1000–4999 g)
- Bulk (5000 g and above)

6. Category Analysis

- Total number of products in each category
- Category-wise revenue estimation

- Highest priced product by category

7. Window Function Analysis

- Calculated cumulative inventory value using SQL Window Functions.
- Useful for identifying products contributing to the majority of inventory value (Pareto/80-20 analysis).

📈 Business Insights Generated

- Categories generating the highest inventory value.
- Products offering the largest discounts.
- High-value products currently out of stock.
- Duplicate product names with multiple SKUs.
- Average pricing across categories.
- Price efficiency (price per gram).
- Inventory contribution using cumulative value analysis.

🚀 Skills Demonstrated

- SQL Query Writing
- Data Cleaning
- Exploratory Data Analysis (EDA)
- Business Analytics
- Inventory Analysis
- Revenue Analysis
- Window Functions
- Conditional Logic
- Data Aggregation

💻 Technologies Used

- MySQL
- MySQL Workbench

📁 Project Structure

```
Zepto-SQL-Analysis/
│
├── dataset/
│   └── zepto_v2.csv
```

```
│
├── sql/
│   └── zepto_analysis.sql
│
├── screenshots/
│   ├── query_results.png
│   └── dashboard.png
│
└── README.md
```

🔮 Future Improvements

- Build an interactive Power BI dashboard.
- Add advanced SQL queries using CTEs and ranking functions.
- Perform customer and sales trend analysis (if transactional data is available).
- Create SQL views and stored procedures for reusable reporting.

👤 Author

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SQL | Power BI | Data Analytics | C++

If you found this project useful, feel free to ⭐ the repository.